

Year 8 Mini-Test 3
Perimeter, Area and Linear graphs



Baldivis
Secondary College

Resources Allowed
Calculator and page of notes

Name: Anne Sims Date: _____ Score: 35/35

1. [5 marks]

Convert the following units

a) $3\text{cm} = \underline{30} \text{ mm}$ ✓

d) $8\text{cm}^2 = \underline{800} \text{ mm}^2$ ✓

b) $8\text{m} = \underline{800} \text{ cm}$ ✓

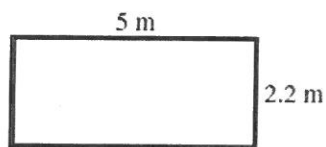
e) $21500 \text{ m}^2 = \underline{0.0215} \text{ km}^2$ ✓

c) $3000\text{m} = \underline{3} \text{ km}$ ✓

2. [10 marks]

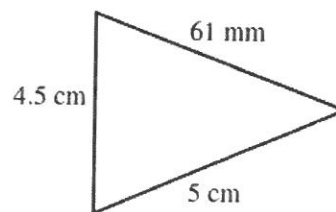
i) Find the perimeter of the following

a)



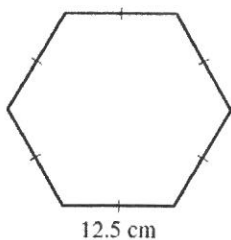
$5 + 5 + 2.2 + 2.2 = \underline{14.4 \text{ m}}$ ✓

b)



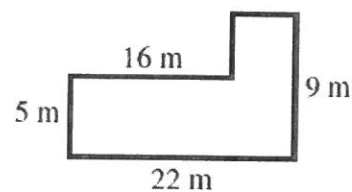
$6.1 + 5 + 4.5 = \underline{15.6 \text{ cm}}$ ✓

c)



$12.5 \times 6 = \underline{75 \text{ cm}}$ ✓

d)



$22 + 22 + 9 + 9 = \underline{62 \text{ m}}$ ✓

i) A square has a perimeter of 12m, what is the length of its sides?

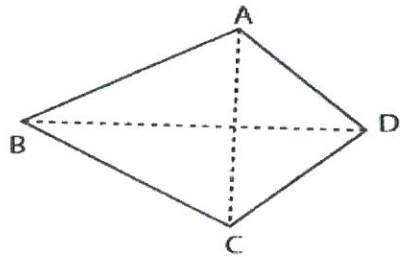


$12 \div 4 = \underline{3 \text{ m}}$ ✓

3. [8 marks]

Find the Area of the following

a)



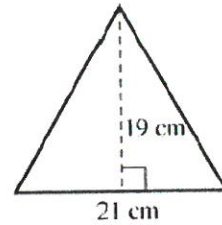
$$\overline{AC} = 5 \text{ cm}$$

$$\overline{BD} = 8 \text{ cm}$$

$$A = 5 \times 8 \div 2$$

$$= 20 \text{ cm}^2$$

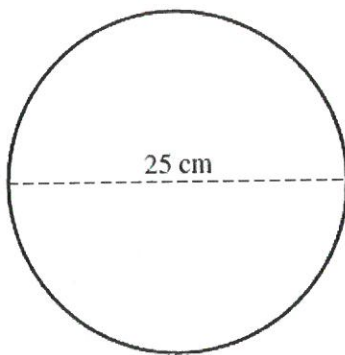
b)



$$A = \frac{1}{2} \times 21 \times 19$$

$$= 199.5 \text{ cm}^2$$

c)

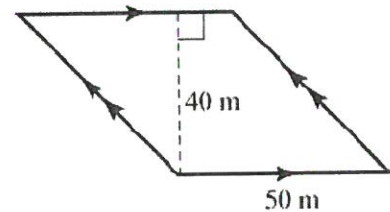


$$A = \pi r^2$$

$$= \pi 25^2$$

$$= 490.87 \text{ cm}^2$$

d)



$$A = 50 \times 40$$

$$= 2000 \text{ m}^2$$

4. [3 marks]

Sally needs to fence her country house block to keep her dog in. The block is a rectangle with length 50m and width 42m.

a) How much fencing will she need to fence her block?

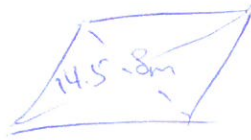
$$50 + 50 + 42 + 42 = 182 \text{ m}$$

b) If it costs \$13 per meter, how much will it cost to fence her block?

$$182 \times 13 = \$2366$$

5. [3 marks]

A landscape gardener charges \$20 per square meter of lawn. A lawn area in the shape of a rhombus and its diagonals are 8m and 14.5m. What would be the cost of laying this lawn?



$$A = 8 \times 14.5 \div 2$$

$$= 58 \text{ m}^2 \checkmark$$

$$\text{Cost} = 20 \times 58 \checkmark$$

$$= \$1160 \checkmark$$

6. [6 marks]

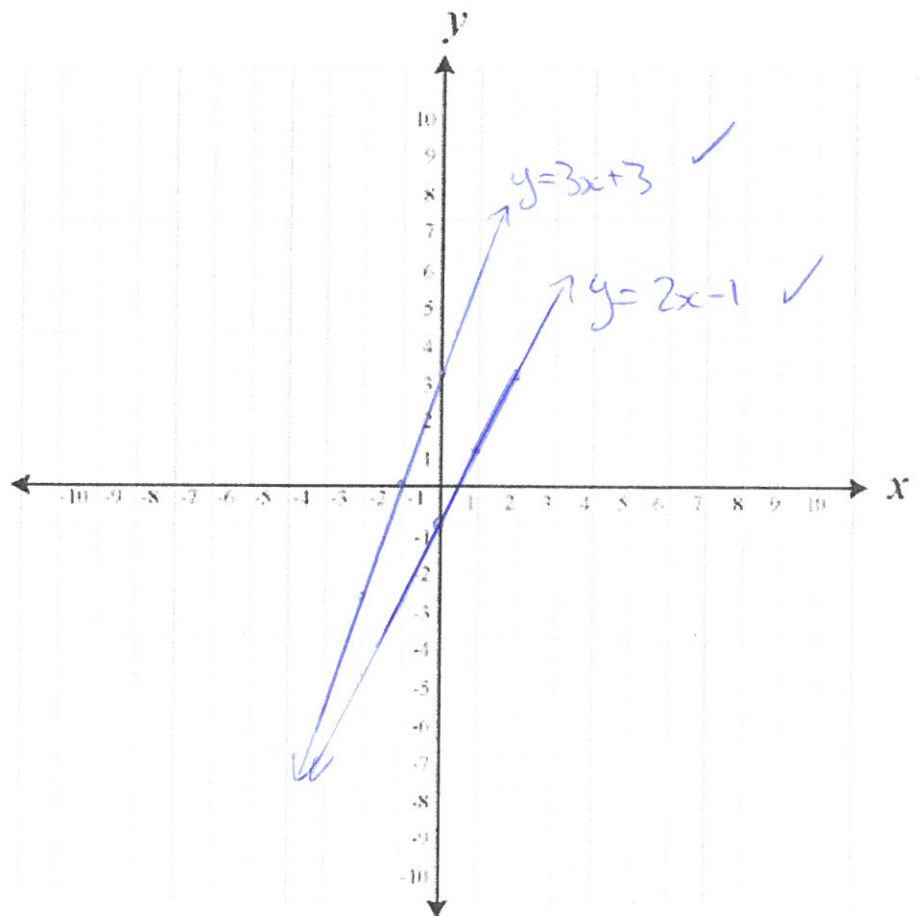
Use the following tables to determine the rule and plot on the Cartesian plane.

x	-2	-1	0	1	2
y	-3	0	3	6	9

Rule: $y = 3x + 3$ ✓

x	-2	-1	0	1	2
y	-5	-3	-1	1	3

Rule: $y = 2x - 1$



END OF TEST

